



1.03 Gallop Agitator System

1. Purpose

The purpose of the Standard Operating Procedure is to identify the hazards and define the steps in operating a Gallop Agitator .

2. Scope

This procedure is intended for use by all Silverpoint employees .

3. Hazard Assessment & Identification

- Personal Injury .
- Equipment Damage
- Equipment Failure /Stress
- Physical Stress -Ergonomics
- Physical Stress -Excessive Noise
- Physical Stress -Reduced or Restricted Visibility
- Mental Stress
- Potential Energy -Pressurized Liquids
- Exposure to Flammable Substance
- Exposure to Toxic/Carcinogenic Substance
- Exposure to Irritating /Sensitizing Substance

4. EHS Requirements

- F/R Coveralls .
- Steel Toed Boots .
- Safety Glasses .
- Personal Monitor
- Gloves .
- Hardhat
- Hearing Protection



- Harness

5. References

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List
- Silverpoint THA 1.03

6. Procedure

- 6.1. Gallop Agitator Operation -The purpose of the agitator is to stir up heavy solids in the bottom of the tank, making it easier to spread .

6.1.1. Engage Agitator PTO – Refer to SOP 1.01 (6.2/6.2.5) for engaging the PTO

6.1.2. Listen for Proper Operation ,you should be able to hear the agitator running , either the agitator motor or fluid moving around inside the tank .

6.1.3. Disengage PTO - Refer to SOP 1.01 (6.2.6).

- 6.2. Gunning With Agitator

6.2.1. Connect Hose for Gunning , find your 4"-2" fitting and 2' nozzle , and then refer to SOP 1.02 on how to properly connect a hose to the truck . Also get the other hose ready (the one that will suck up the shale).

6.2.2. Close your Agitator Isolation Valve. Engage Agitator -Refer to SOP 1.01 (6.2/6.2.5) for engaging the PTO.

6.2.3. Before you open the valve , it is good to tie the end of the hose down or ask someone to help you , so you have control of the hose at all times . When you are satisfied and ready , open the valve .

6.2.4. When you are gunning the product , it really works well if you can use flock water or mud from the rig (ask before using either). When gunning , some trucks work better with 2 or 3 cubes of fluid in the tank , other trucks it doesn't matter . If you swirl the fluid , you are jetting , it seems to break up the shale better . Sometimes volume works better then lots of pressure .

6.2.5. When you are loaded , you can close the agitator valve , this will not hurt anything because the set up allows for circulation in the tank with the outside valve closed and close the load valve .

6.2.6. Refer to SOP 1.02 on the proper procedure for disconnecting your hoses .

6.2.7. Refer to SOP 1.01 (6.2.6), on the proper procedure for disengaging the PTO.

- 6.3. Agitator Troubleshooting

6.3.1. Engage PTO refer to the SOP 1.01 (6.2/6.2.5)



- 6.3.2. If you are going to check the agitator, you will need to back to a designated area where you are able to open the back door. This is usually at the remote sump. Referring to SOP 1.06 (6.2/6.3) may help you.
- 6.3.3. First, you need to test the atmosphere in the tank, for the proper procedure to do this refer to SOP 4.04,
- 6.3.4. After you have done your confined space entry, you will need to open the back door. With the truck on vacuum build a 20" Vacuum, spin the wing nuts off, bleed off the vacuum by slowly moving the 4-way valve into neutral, open the back door. **Make sure that you install the safety bar.**
- 6.3.5. With your PTO still running, engage your agitator, you can refer to SOP 1.08 (6.2.1/6.2.2) to help you do this procedure properly.
- 6.3.6. Listen to the operation of the agitator; if there is something wrong with it you will hear it right away. You can also visually inspect the basket to see if anything is caught in it or plugging the screen. Feel each jet to make sure that there is good air flow, the front one will have a little less than the back because it is farther away from the agitator.
- 6.3.7. Remember that these are fluid agitators, and they perform the best when moving fluid, so as long as you have air flow it is working. If you have a plugged jet, see if you can unplug it, if not call your supervisor. Even with a plugged jet or two, the agitator will still work, just not as efficient.
- 6.3.8. Check to make sure the isolation valve and ram is mounted good and not loose (jam nut is tight), check to make sure that there is no rub marks on hydraulic hoses, or oil leaking anywhere, If you see traces of oil you need to shut down the agitator immediately, try and find where it is leaking from before calling in, it just makes it easier to access on what action needs to take place as far as mechanical goes.
- 6.3.9. If there are no repairs and everything seems to be ok, remove your safety bar and close your back door. Once again with your truck on vacuum (to help suck the door tight), tighten the wing nuts, **you don't need a hammer, just as tight as you can get them.**
- 6.3.10. Disengage Agitator and PTO; refer to the SOP 1.08 (6.2.1/6.2.2) to help you with this procedure.

6.4. Parts of the agitator

- 6.4.1. Hydraulic pump, Orbit Motor, Impeller, Piping, Jets, Bearing Block, Atmospheric Chamber, Grease Nipples, PTO, Valve Controls & Agitator Isolation Valve.

6.5. Regular Maintenance is very important to having an Agitator work well for you.

- 6.5.1. Checking the Atmospheric Chamber on the side of the tank for contamination is very important (this is usually a ¼ inch valve). If Oil or Mud comes out when you open it, call your supervisor or the shop. This should be checked on a daily basis.
- 6.5.2. Usually beside the Atmospheric Chamber Valve, there is a Grease Nipple for the Agitator, some of the Trucks have two, and it should be greased on a daily basis and no longer can be over greased. **20 pumps a day min.** You can grease with the Agitator running.
- 6.5.3. **DO NOT** run with the Agitator down the road, the fluctuation in the rpms, do to shifting, will **SERIOUSLY**



- damage the components of the agitator. Even with the automatic transmissions, you **CAN NOT** keep your RPMS under control enough to run with the Agitator on, they shift at 1500 RPM or higher.
- 6.5.4. Make sure that you follow the Land Spreading Procedure (1.08) and suck out your Agitator Valves. It is always good to keep the Lines and Valves clear of Mud and Cement, so your Agitator keeps working properly.
- 6.5.5. If your Agitator is plugged or simply not working, and you haven't lost your Oil in your load, you can still spread the load. If you follow the Burping Procedure (1.05), this agitates the load enough to allow you to spread it (this is the old school method). In some cases, you might even have to stop halfway through the load and burp it again, it WILL work, this is how it was done before Agitators were around.
- 6.5.6. To clean out the Agitator Line, spin the Impeller and suck out the line. This **MUST** be done after every cement job, and after heavy shale. Spinning the agitator will clean the impeller while the 4-way is on vacuum. **To properly clean the Agitator Valve, Agitator Line (inside the tank) and the Jet line, build up to -20 psi and open the Agitator valve. Repeat multiple times if needed.**

Remember : if you are in doubt about what to do, call your supervisor